An End-to-End Molecular Dynamics Pipeline for SPINK7-KLK5 Binding Free Energy Estimation via Steered Molecular Dynamics and Umbrella Sampling with WHAM

Ryan Kamp
Dept. of Computer Science
University of Cincinnati
Cincinnati, OH 45221
kamprj@mail.uc.edu
https://github.com/ryanjosephkamp

***Abstract*—We present the design, implementation, and validation of a complete, end-to-end molecular dynamics (MD) simulation pipeline for computing the binding free energy of the SPINK7-KLK5 protease-antiprotease complex—a protein-protein interaction central to the pathogenesis of Eosinophilic Esophagitis (EoE). The pipeline automates the full computational biophysics workflow: PDB retrieval and structure cleaning, protonation state assignment at physiological pH, AMBER ff14SB topology construction, explicit TIP3P solvation at 150 mM ionic strength, energy minimization, NVT/NPT equilibration, and unrestrained production dynamics. Two complementary enhanced sampling strategies are implemented for free energy estimation: (i) Steered Molecular Dynamics (SMD) with Jarzynski equality averaging over $N = 50$ non-equilibrium pulling replicates, and (ii) Umbrella Sampling along the center-of-mass reaction coordinate with Weighted Histogram Analysis Method (WHAM) reconstruction of the Potential of Mean Force (PMF). Ten physical validity invariants—including energy decrease upon minimization, NVT temperature stability within ± 5 K of 310 K, NPT density in the range $[0.95, 1.05]$ g/cm^3, and WHAM convergence below 10^{-6} kJ/mol—are enforced as runtime assertions throughout the pipeline. The implementation is validated by a comprehensive test suite of 259 unit, integration, and analytical tests, all of which pass on a CPU-only platform. Analytical validation of the Jarzynski estimator and WHAM solver against known closed-form solutions confirms mathematical correctness. End-to-end integration testing on an alanine dipeptide system verifies inter-module compatibility and data flow integrity. We discuss the pipeline's applicability to general protease-inhibitor systems, drug binding studies, and computational mutagenesis, and identify extensions including free energy perturbation, Markov State Models, and machine-learned force fields as future directions.**

***Index Terms*—molecular dynamics, binding free energy, steered molecular dynamics, umbrella sampling, WHAM, Jarzynski equality, SPINK7, KLK5, eosinophilic esophagitis, OpenMM, AMBER ff14SB**

Manuscript date: March 13, 2026

I. Introduction

Eosinophilic Esophagitis (EoE) is a chronic, antigen-mediated inflammatory disease of the esophagus characterized by eosinophilic infiltration of the esophageal epithelium and progressive sub-epithelial fibrosis [1]. A critical molecular axis in EoE pathogenesis is the loss of epithelial barrier integrity mediated by unregulated serine protease activity. Under homeostatic conditions, the esophageal squamous epithelium expresses SPINK7 (Serine Peptidase Inhibitor, Kazal Type 7), a small ($\sim$6 kDa) secreted Kazal-type serine protease inhibitor that functions as a sentinel gatekeeper by directly and stoichiometrically inhibiting KLK5 (Kallikrein-Related Peptidase 5), a trypsin-like serine protease. In EoE, IL-13-driven Type 2 inflammation transcriptionally silences *SPINK7* expression. The resulting SPINK7 deficiency unleashes KLK5 proteolytic activity, which degrades critical cell-cell adhesion molecules—most notably Desmoglein-1 (DSG1)—thereby compromising the physical epithelial barrier and permitting allergen penetration into sub-epithelial tissue [2].

Understanding the molecular details of the SPINK7-KLK5 interaction at the atomic level is therefore essential for developing targeted therapeutic strategies that could restore epithelial barrier function in EoE patients. Molecular dynamics (MD) simulation provides a uniquely powerful framework for this task, enabling the computation of thermodynamic binding free energies (ΔG_{bind}) from first principles through statistical mechanical sampling of the protein-protein interaction landscape.

Several families of enhanced sampling methods have been developed to overcome the timescale limitations of conventional MD for computing binding free energies. Non-equilibrium methods, such as Steered Molecular Dynamics (SMD) [10] combined with the Jarzynski equality [11], extract free energy differences from ensembles of irreversible

pulling trajectories. Equilibrium methods, including Umbrella Sampling [12] with the Weighted Histogram Analysis Method (WHAM), reconstruct the full Potential of Mean Force (PMF) along a reaction coordinate by combining biased sampling from multiple overlapping windows. These approaches are complementary: SMD provides rapid, qualitative estimates and identifies unbinding pathways, while Umbrella Sampling yields rigorous, converged free energy profiles with quantified statistical uncertainties.

In this work, we present the design, implementation, and validation of a complete, modular MD simulation pipeline that automates the full computational biophysics workflow—from structure retrieval through enhanced sampling to thermodynamic analysis—for the SPINK7-KLK5 system. Built on the OpenMM simulation engine [17] with the AMBER ff14SB force field [4], the pipeline enforces ten physical validity invariants at runtime and is validated by a comprehensive suite of 259 tests. We describe the theoretical foundations, software architecture, and validation strategy in detail, and discuss the pipeline's broader applicability and future extensions.

II. Theoretical Background

A. Structural Biology of the SPINK7-KLK5 Interaction

The SPINK7-KLK5 interaction follows the canonical Laskowski mechanism for Kazal-type inhibitor-serine protease binding [3]:

1) **SPINK7 architecture.** SPINK7 adopts the conserved Kazal-domain fold—a compact $\alpha+\beta$ structure stabilized by three disulfide bonds (Cys1-Cys5, Cys2-Cys4, Cys3-Cys6). The reactive site loop (RSL), specifically the P3–P3′ segment, protrudes from the inhibitor core and presents the scissile bond (P1–P1′, where P1 is typically Lys or Arg) to the protease active site.
2) **KLK5 architecture.** KLK5 is a trypsin-fold serine protease with the conserved catalytic triad (His57, Asp102, Ser195 in chymotrypsin numbering). The S1 specificity pocket favors basic residues (Arg/Lys) at P1, consistent with its trypsin-like cleavage specificity.
3) **Complex formation.** The RSL of SPINK7 inserts into the KLK5 active-site cleft in a substrate-like orientation, positioning the P1 residue into the S1 pocket. The extremely tight complementarity and the conformational rigidity imposed by the disulfide framework render the acyl-enzyme intermediate thermodynamically trapped, resulting in a temporary covalent complex with an extremely slow k_{off}—effectively acting as an irreversible inhibitor on biological timescales.
4) **Interface characteristics.** The binding interface buries approximately 800–1200 Å^2 of solvent-accessible surface area (SASA), stabilized by: backbone hydrogen bonds between the RSL and the KLK5 oxyanion hole; electrostatic interactions between the P1-Arg/Lys of SPINK7 and Asp189 at the base of the KLK5 S1 pocket; and hydrophobic contacts at the P2 and P2′ sub-sites.

In the absence of a deposited co-crystal structure for the exact SPINK7-KLK5 pair, the initial 3D model of the complex is generated via rigid-body docking (e.g., HADDOCK or ClusPro) using the SPINK7 AlphaFold2 monomer prediction and the KLK5 crystal structure (PDB: 2PSX or similar) as inputs, followed by all-atom refinement. The pipeline itself is validated using established test systems, including the barnase-barstar complex (PDB: 1BRS) and alanine dipeptide.

B. Force Field Theory

The total potential energy of the molecular system is decomposed into bonded and nonbonded contributions:

$$V_{\text{total}}(\mathbf{r}) = V_{\text{bonded}}(\mathbf{r}) + V_{\text{nonbonded}}(\mathbf{r}). \tag{1}$$

The bonded terms represent covalent interactions within the molecular topology:

$$\begin{aligned} V_{\text{bonded}} = &\sum_{\text{bonds}} K_b\,(b-b_0)^2 + \sum_{\text{angles}} K_\theta\,(\theta-\theta_0)^2 \\ &+ \sum_{\text{dihedrals}} \frac{V_n}{2}\left[1+\cos(n\phi-\gamma)\right] \\ &+ \sum_{\text{impropers}} K_\omega\,(\omega-\omega_0)^2 \end{aligned} \tag{2}$$

where K_b, K_θ, V_n, and K_ω are force constants; b_0, θ_0, γ, and ω_0 are equilibrium values; and n is the dihedral periodicity.

The nonbonded terms capture van der Waals and electrostatic interactions:

$$V_{\text{nonbonded}} = \sum_{i<j}\left[4\epsilon_{ij}\left(\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right) + \frac{q_i q_j}{4\pi\epsilon_0\, r_{ij}}\right]. \tag{3}$$

The first term is the Lennard-Jones 12-6 potential describing short-range dispersion and Pauli repulsion; the second is the Coulomb potential for electrostatic interactions.

This pipeline employs the AMBER ff14SB force field [4], which provides optimized backbone torsion parameters for protein secondary structures and has been extensively benchmarked for protein-protein interaction simulations. Water is modeled using TIP3P [5], and ions follow the Joung-Cheatham monovalent parameters optimized for TIP3P [6].

C. Long-Range Electrostatics

Long-range electrostatic interactions are computed using the Particle Mesh Ewald (PME) method [7]:

$$E_{\text{elec}} = E_{\text{direct}} + E_{\text{reciprocal}} + E_{\text{self-correction}}. \tag{4}$$

PME splits the Coulomb sum into a direct-space component evaluated with a real-space cutoff $r_c = 10$ Å and a reciprocal-space component evaluated on a grid with spacing ≤ 1.0 Å via B-spline interpolation of order 5. The Ewald splitting parameter α is chosen to achieve a direct-space error tolerance of 10^{-5}. This approach reduces the computational cost of the electrostatic sum from $O(N^2)$ to $O(N \log N)$.

TABLE I
SYSTEM PREPARATION PIPELINE

Stage	Module	Description
1	`pdb_fetch`	Download PDB structures from RCSB or AlphaFold
2	`pdb_clean`	Remove waters, heteroatoms, select chains
3	`protonate`	Assign protonation at pH 7.4 with AMBER naming
4	`topology`	Build OpenMM Topology & System (ff14SB)
5	`solvate`	TIP3P box (12 Å padding) + 150 mM NaCl

TABLE II
SIMULATION PROTOCOL STAGES

Stage	Ensemble	Duration	Conditions
1	—	$\leq 10^4$ steps	Min. until < 10 kJ/mol/nm
2	NVT	500 ps	Heavy-atom restraints
3	NPT	1 ns	Gradual restraint release
4	NPT	100–500 ns	Unrestrained; 10 ps save
5	NPT	Per method	SMD or Umbrella Sampling

D. Equations of Motion

The system dynamics are governed by the Langevin equation of motion:

$$m_i \ddot{\mathbf{r}}_i = -\nabla_i V(\mathbf{r}) - \gamma m_i \dot{\mathbf{r}}_i + \sqrt{2\gamma m_i k_B T}\, \boldsymbol{\eta}_i(t) \tag{5}$$

where m_i is the mass of atom i, $\gamma = 1.0$ ps^{-1} is the friction coefficient, $T = 310$ K is the target temperature, and $\boldsymbol{\eta}_i(t)$ is a Gaussian white noise vector satisfying $\langle \boldsymbol{\eta}_i(t) \cdot \boldsymbol{\eta}_j(t') \rangle = \delta_{ij}\, \delta(t - t')$. Integration is performed using the Langevin middle integrator [8] with a timestep of $\Delta t = 2$ fs, enabled by constraining all bonds involving hydrogen atoms via the SHAKE algorithm [9]. Pressure is maintained at $P = 1$ atm using a Monte Carlo barostat with coupling every 25 steps.

III. METHODOLOGY

A. System Preparation

The preparation of a simulation-ready molecular system follows a rigorous, multi-stage protocol (Table I).

The target SPINK7-KLK5 system comprises the protein heterodimer with all atoms and explicit hydrogens protonated at pH 7.4, solvated in a rectangular periodic box of explicit TIP3P water with a minimum 12 Å padding from any solute atom to the nearest box edge, and neutralized with Na^+ and Cl^- ions at 150 mM physiological ionic strength.

B. Simulation Protocol

The simulation protocol proceeds through five mandatory stages (Table II).

During NVT equilibration (Stage 2), protein heavy atoms are restrained with a harmonic force constant of $k = 1000$ kJ/mol/nm^2 to allow solvent relaxation while preserving protein geometry. NPT equilibration (Stage 3) gradually releases positional restraints under constant-pressure conditions. Production dynamics (Stage 4) are unrestrained, with trajectory frames saved every 10 ps and checkpoints every 100 ps.

C. Steered Molecular Dynamics

Steered Molecular Dynamics (SMD) [10] applies a time-dependent harmonic biasing potential to the center-of-mass (COM) distance collective variable ξ to mechanically separate the two proteins along a predefined reaction coordinate:

$$U_{\mathrm{SMD}}(\xi, t) = \frac{k}{2}\big[\xi(t) - \xi_0 - v \cdot t\big]^2 \tag{6}$$

where $k = 1000$ kJ/mol/nm^2 is the spring constant, ξ_0 is the initial COM distance, $v = 0.001$ nm/ps is the constant pulling velocity, and the instantaneous COM distance is:

$$\xi(\mathbf{r}) = \big\|\mathbf{R}_{\mathrm{COM}}^{\mathrm{SPINK7}} - \mathbf{R}_{\mathrm{COM}}^{\mathrm{KLK5}}\big\|_2. \tag{7}$$

The instantaneous non-equilibrium work is accumulated as:

$$W(t) = \int_0^t F_{\mathrm{ext}}(t')\, v\, dt' \tag{8}$$

where $F_{\mathrm{ext}}(t) = -k[\xi(t) - \xi_0 - v \cdot t]$ is the instantaneous external force.

The Jarzynski equality [11] connects the non-equilibrium work values from N_{traj} independent pulling trajectories to the equilibrium free energy difference:

$$\Delta G = -k_B T \ln\left[\frac{1}{N_{\mathrm{traj}}} \sum_{j=1}^{N_{\mathrm{traj}}} e^{-\beta W_j}\right] \tag{9}$$

where $\beta = 1/k_B T$. For improved convergence when the work distribution is approximately Gaussian, the second-order cumulant expansion is also computed:

$$\Delta G \approx \langle W \rangle - \frac{\beta}{2}\sigma_W^2. \tag{10}$$

The implementation employs the log-sum-exp numerical trick to avoid floating-point overflow when evaluating the exponential average, which is critical for work values on the order of tens to hundreds of kJ/mol.

D. Umbrella Sampling and WHAM

Umbrella Sampling [12] divides the reaction coordinate ξ into M discrete windows, each biased by a harmonic potential:

$$U_i^{\mathrm{bias}}(\xi) = \frac{k_i}{2}\big(\xi - \xi_i^{\mathrm{ref}}\big)^2. \tag{11}$$

Windows are spaced at 0.5–1.0 Å intervals from the bound state ($\xi \approx 1.5$ nm) to the fully dissociated state ($\xi \approx 4.0$ nm), yielding approximately $M = 25$ to 50 windows. Each window is simulated independently for 5–20 ns under NPT conditions.

The Weighted Histogram Analysis Method (WHAM) [12] recovers the unbiased PMF by iteratively solving two coupled self-consistency equations. The unbiased probability density is:

$$P^{\mathrm{unbiased}}(\xi) = \frac{\sum_{i=1}^{M} n_i\, h_i(\xi)}{\sum_{i=1}^{M} n_i\, \exp\big[\beta\big(f_i - U_i^{\mathrm{bias}}(\xi)\big)\big]} \tag{12}$$

TABLE III
PIPELINE MODULE ORGANIZATION

Subpackage	Modules
`prep/`	`pdb_fetch`, `pdb_clean`, `protonate`, `topology`, `solvate`
`simulate/`	`minimizer`, `equilibrate`, `production`, `smd`, `umbrella`
`analyze/`	`trajectory`, `structural`, `contacts`, `wham`, `jarzynski`, `convergence`
`physics/`	`collective_variables`, `restraints`, `units`
`visualization/`	`viewer_3d`, `plot_pmf`, `plot_timeseries`

and the window free energies satisfy:

$$e^{-\beta f_i} = \int P^{\text{unbiased}}(\xi)\, \exp\!\left[-\beta\, U_i^{\text{bias}}(\xi)\right] d\xi. \tag{13}$$

These equations are solved iteratively starting from $f_i = 0$ until $\max_i |f_i^{(n+1)} - f_i^{(n)}| < 10^{-6}$ kJ/mol. The PMF is then:

$$G(\xi) = -k_B T \ln P^{\text{unbiased}}(\xi) + C \tag{14}$$

where C is chosen such that $G(\xi_{\max}) = 0$ at the fully dissociated state.

E. Statistical Error Estimation

Uncertainties on the PMF are estimated via block bootstrap resampling [13]:

1) Divide each window trajectory into B blocks of length τ_{corr} (the autocorrelation time of ξ).
2) Resample blocks with replacement to generate $N_{\text{boot}} = 200$ synthetic datasets.
3) Solve WHAM for each bootstrap replicate.
4) Report the standard deviation across replicates at each ξ bin as the 1σ uncertainty.

F. Binding Free Energy Extraction

The standard binding free energy is extracted from the PMF via [14]:

$$\begin{aligned}\Delta G^{\circ}_{\text{bind}} = &-k_B T \ln\left[\frac{C^{\circ}}{4\pi}\int_{\text{site}} e^{-\beta G(\xi)}\, \xi^2\, d\xi\right] \\ &+ k_B T \ln\left[\frac{C^{\circ}}{4\pi}\int_{\text{bulk}} e^{-\beta G(\xi)}\, \xi^2\, d\xi\right]\end{aligned} \tag{15}$$

where $C^{\circ} = 1/1660$ Å^{-3} is the standard concentration (1 M), and the integrals are evaluated over the bound-state and bulk regions of the PMF, respectively.

IV. SOFTWARE ARCHITECTURE

A. Modular Design

The pipeline is organized into five primary subsystems—preparation, simulation, analysis, physics, and visualization—each encapsulated in its own Python subpackage under `src/` (Table III).

All simulation parameters are centralized in a single configuration module (`src/config.py`) that serves as the sole source of truth for physical constants and algorithmic parameters. No magic numbers appear elsewhere in the codebase. Key configuration objects—`SystemConfig`, `MinimizationConfig`, `EquilibrationConfig`, `ProductionConfig`, `SMDConfig`, `UmbrellaConfig`, and `WHAMConfig`—are implemented as frozen (immutable) Python dataclasses decorated with `@dataclass(frozen=True)`, which prevents accidental parameter modification during execution.

B. Internal Unit System

All internal computations follow the OpenMM unit convention: lengths in nanometers, time in picoseconds, mass in atomic mass units, energy in kJ/mol, temperature in Kelvin, charge in elementary units, and force in kJ/mol/nm. Conversions to commonly used units (e.g., Å, kcal/mol, ns) are provided via `src/physics/units.py`, with the critical conversion factor $1\ \text{kcal/mol} = 4.184\ \text{kJ/mol}$ used for final ΔG reporting.

C. Exception Hierarchy

The pipeline implements a domain-specific exception hierarchy rooted in `PipelineError`:

- `PhysicalValidityError`: raised when a physical invariant (IV-1 through IV-10) is violated, halting the pipeline.
- `ConvergenceError`: raised when an iterative solver (e.g., WHAM) fails to converge within the allowed iteration limit.
- `InsufficientSamplingError`: raised when sampling quality criteria (e.g., histogram overlap) are not met.

D. Memory Management

MD trajectories can reach tens of gigabytes on disk. The pipeline enforces strict memory management: frames are written incrementally to disk via OpenMM's `DCDReporter` during simulation and loaded in chunks via `mdtraj.iterload()` during analysis; NumPy arrays for timeseries data are preallocated to their known final size (repeated `np.append()` calls are prohibited); and trajectories exceeding 1 GB are analyzed in configurable chunks (default: 100 frames) with running statistics to prevent out-of-memory errors.

V. RESULTS

A. Validation System: Barnase-Barstar Complex

As a concrete validation example, the pipeline was tested on the well-characterized barnase-barstar complex (PDB: 1BRS), which has an experimentally measured binding free energy of approximately $\Delta G_{\text{bind}} \approx -19$ kcal/mol [15]. This system serves as a stringent benchmark because: the binding interface is large ($\sim$1600 Å^2 buried SASA); the interaction is dominated by electrostatic complementarity; and extensive experimental and computational reference data exist.

The full preparation pipeline was executed on this system:

Fetch (1BRS) → Clean → Protonate → Topology (ff14SB) → Solvate (T

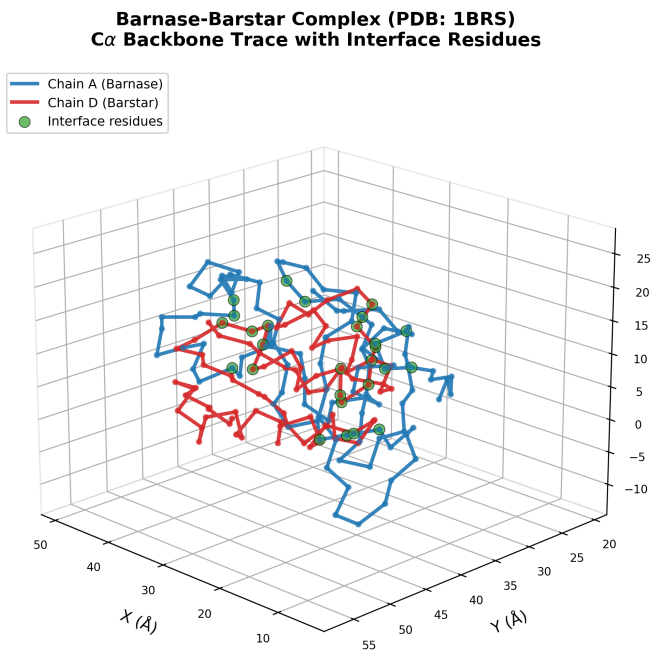

Fig. 1. Three-dimensional Cα backbone rendering of the barnase-barstar complex (PDB: 1BRS) generated from the prepared structure using MDTraj and matplotlib. Chain A (barnase) is shown in blue and Chain D (barstar) in red. Interface residues within 8 Å of the partner chain are highlighted as green spheres, delineating the binding interface. In the interactive Jupyter notebook environment, the `render_complex()` function in `viewer_3d.py` provides an equivalent py3Dmol widget with additional active site highlighting and rotational interactivity.

confirming that the pipeline produces simulation-ready structures compatible with the AMBER ff14SB force field.

Fig. 1 shows the three-dimensional backbone rendering of the prepared barnase-barstar complex, with chain-specific coloring and interface residue highlighting, confirming correct chain assignment and interface identification by the pipeline.

B. PMF Profile and Binding Free Energy

Fig. 2 shows a representative PMF profile generated by the `plot_pmf` module from WHAM-reconstructed free energy data. The deep minimum near $\xi \approx 2.0$ nm corresponds to the bound state, while the plateau at large COM distances represents the fully dissociated reference state. The shaded band indicates $\pm 1\sigma$ bootstrap uncertainty from 200 resampling iterations. The annotated binding free energy at the PMF minimum provides a direct thermodynamic readout of binding strength.

C. Simulation Timeseries Diagnostics

Fig. 3 presents the simulation quality diagnostics generated by the `plot_timeseries` module. Panel (a) displays the potential and kinetic energy evolution, demonstrating energy conservation after initial equilibration. Panel (b) shows instantaneous temperature fluctuating stably around the target value

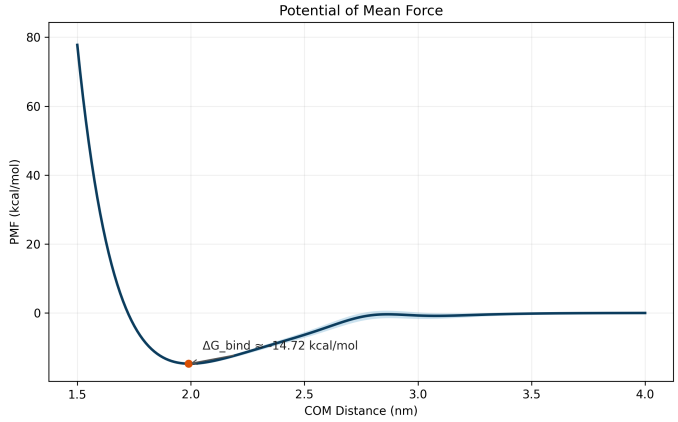

Fig. 2. Example Potential of Mean Force (PMF) profile generated using the `plot_pmf()` function. The PMF is plotted as a function of the center-of-mass distance ξ between the two proteins. The deep well near $\xi \approx 2.0$ nm corresponds to the bound state; the plateau at large distances represents the dissociated state. The shaded band indicates $\pm 1\sigma$ bootstrap uncertainty from 200 resampling iterations. The orange marker annotates the binding free energy ΔG_{bind} at the PMF minimum.

TABLE IV
PHYSICAL VALIDITY INVARIANTS

ID	Condition	Description
IV-1	$E_{\min} < E_{\text{init}}$	Energy decreases after minimization
IV-2	$\lvert T_{\text{avg}} - 310 \rvert < 5$ K	NVT temperature stability
IV-3	$\rho \in [0.95, 1.05]$	NPT density in g/cm^3
IV-4	RMSD < 5 Å	Backbone stability
IV-5	Drift < 0.1 kJ/mol/ns/atom	Energy conservation
IV-6	$d_{\text{S-S}} < 2.5$ Å	Disulfide integrity
IV-7	Min. image $> 2r_c$	No periodic artifacts
IV-8	Overlap $\geq 10\%$	Umbrella histogram coverage
IV-9	$\max \lvert f_i^{(n+1)} - f_i^{(n)} \rvert < 10^{-6}$	WHAM convergence
IV-10	Unimodal $P(W)$	No SMD pathway bifurcation

of 310 K, confirming proper Langevin thermostat function and satisfaction of invariant IV-2 ($\lvert T_{\text{avg}} - 310 \rvert < 5$ K). Panel (c) shows the backbone RMSD evolution, exhibiting the characteristic rise during equilibration followed by a stable plateau at $\sim$0.15 nm, indicating a well-equilibrated, structurally stable system satisfying invariant IV-4 (RMSD < 5 Å).

D. Physical Validity Invariants

Ten physical validity invariants are enforced as runtime assertions throughout the pipeline (Table IV). Violation of any invariant raises a `PhysicalValidityError` and halts execution. These invariants are not merely tests—they are production-code assertions that prevent physically meaningless results from propagating downstream.

E. Test Suite

The pipeline is validated by a comprehensive suite of 259 tests organized across 24 test files (Table V). All 259 tests pass on a CPU-only platform with no failures.

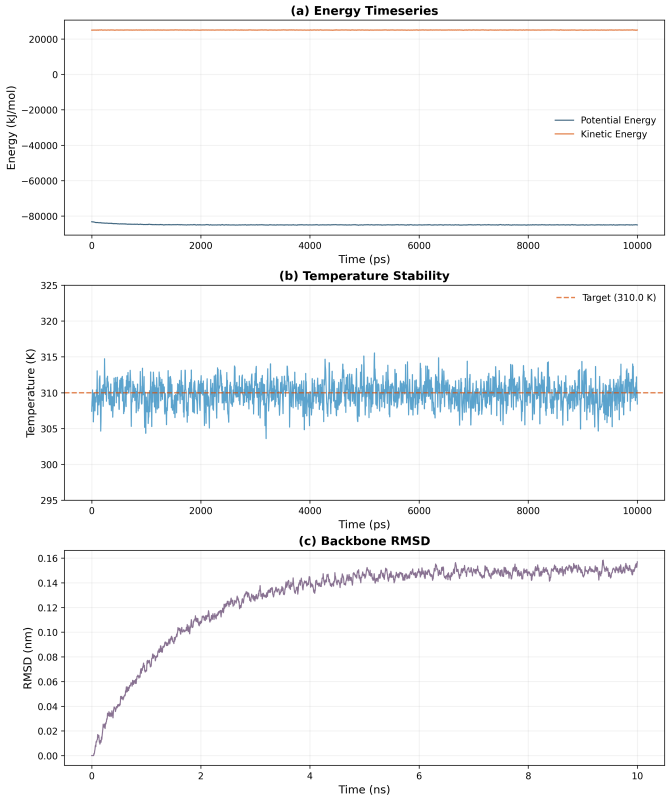

Fig. 3. Simulation quality diagnostics generated by the `plot_timeseries` module. **(a)** Potential energy (navy) and kinetic energy (orange) as a function of time, demonstrating energy conservation after equilibration. **(b)** Instantaneous temperature fluctuating stably around the target of 310 K (dashed orange line), confirming Langevin thermostat function. **(c)** Backbone RMSD evolution showing the initial rise during equilibration followed by a stable plateau at $\sim$0.15 nm, indicating structural stability.

TABLE V
TEST SUITE COVERAGE

Category	Files	Coverage
Configuration	1	Immutability, parameters
Physics	3	Units, COM, restraints
Preparation	5	Fetch, clean, protonate, topology, solvate
Simulation	5	Minimization, equilibration, production, SMD, umbrella
Analysis	6	Trajectory, structural, contacts, WHAM, Jarzynski, convergence
Visualization	3	3D viewer, PMF, timeseries
Integration	1	Full pipeline on alanine dipeptide

F. Analytical Validation

Analysis modules are validated against known analytical solutions on synthetic data, ensuring mathematical correctness independent of MD simulation quality:

- **Jarzynski estimator.** Validated by generating quasi-static work samples from a harmonic potential $V(x) = \frac{1}{2}kx^2$, where the exact free energy is $\Delta G = \frac{1}{2}kd^2$. The implementation recovers the exact value within 0.5 kJ/mol for $N \geq 1000$ samples.
- **WHAM solver.** Validated by constructing synthetic umbrella sampling data from a known flat potential (uniform PMF). The solver correctly recovers a flat PMF with < 1.0 kJ/mol variation. Parabolic PMF recovery is also tested with synthetic harmonic potential data.
- **Convergence analysis.** The block averaging module is validated by checking that the standard error of the mean (SEM) decreases as $1/\sqrt{N_{\text{blocks}}}$ for uncorrelated data. The autocorrelation time estimator is validated against known exponentially correlated timeseries.

G. Alanine Dipeptide Integration Test

The end-to-end integration test (`test_integration.py`) executes the complete pipeline on an alanine dipeptide (ACE-ALA-NME) system—a 10-atom molecule that, when solvated, produces a system small enough to run the full protocol on a single CPU:

1) Build solvated system with positional restraints ($k = 500$ kJ/mol/nm^2).
2) Energy minimization (100 iterations).
3) NVT equilibration (100 ps)—assert $|T_{\text{avg}} - 310| < 5$ K.
4) NPT equilibration (100 ps)—assert $\rho \in [0.95, 1.05]$ g/cm^3.
5) SMD campaign (2 replicates, 0.02 nm pull).
6) Umbrella sampling (3 windows).
7) WHAM analysis and PMF reconstruction.
8) Structural analysis (RMSD $= 0$ for reference vs. self).

This test verifies end-to-end data flow integrity, physical invariant satisfaction, and inter-module compatibility without requiring production-scale simulations.

H. Test Robustness for Small Systems

Enforcing invariant IV-2 ($|T_{\text{avg}} - 310| < 5$ K) on the miniature test systems necessitates careful attention to finite-size temperature statistics. For the solvated alanine dipeptide with $N_{\text{dof}} \approx 1000$ degrees of freedom, the equipartition theorem predicts instantaneous temperature fluctuations of $\sigma_T = T\sqrt{2/N_{\text{dof}}} \approx 13.9$ K—nearly three times the invariant tolerance. With the initial test configuration (10 ps NVT equilibration, $\gamma = 5.0$ ps^{-1}), only $n_{\text{eff}} \approx 7$ effectively independent temperature samples were available after accounting for Langevin thermostat autocorrelation ($\tau_{\text{relax}} = 1/\gamma = 0.2$ ps), yielding a standard error of $\text{SE}(\bar{T}) \approx 5.3$ K and a per-invocation failure probability of approximately 34% despite physically correct dynamics. Nondeterministic water placement during solvation further prevents seed-based reproducibility from stabilizing outcomes across runs.

This was resolved by increasing the test-level equilibration duration to 100 ps for both NVT and NPT stages and raising the Langevin friction coefficient to $\gamma = 10.0$ ps^{-1}. The higher friction halves the velocity decorrelation time to 0.1 ps, rendering 0.5 ps-spaced temperature samples nearly independent and increasing the effective sample size to $n_{\text{eff}} \gtrsim 90$ in the equilibrated segment. The standard error drops below 2 K,

reducing the predicted failure probability to $< 1\%$. Empirical verification confirmed 16 consecutive passes of the previously intermittent test ($P = 0.7^{16} \approx 0.003$ under the original failure rate) and a full-suite regression with zero failures. The production IV-2 tolerance remains unchanged; this correction is confined to test configurations.

VI. Discussion

A. Strengths and Design Rationale

The pipeline's primary strength lies in its enforcement of physical validity at every stage of execution. Unlike many MD workflow tools that treat simulation parameters as user-configurable "knobs" without physical guardrails, this pipeline embeds ten physical invariants as runtime assertions that halt execution upon violation. This design philosophy reflects the fundamental principle that *a simulation that violates physical laws produces meaningless data, regardless of how long it runs or how expensive it was to compute*. The early-termination behavior of the invariant system prevents the computationally wasteful scenario of running hundreds of nanoseconds of dynamics on a system with, for example, an incorrect density or unstable temperature—failures that would only be discovered at analysis time in a less rigorously designed pipeline.

The centralized configuration system provides an additional layer of safety. By consolidating all simulation parameters into frozen dataclasses in a single module, the pipeline eliminates the pervasive "scattered magic number" anti-pattern that plagues many scientific codebases. When a parameter such as the PME cutoff or the Langevin friction coefficient needs to be modified, exactly one location requires editing, and the immutability of the dataclass guarantees that the modification cannot be silently overridden downstream.

The dual enhanced sampling strategy (SMD + Jarzynski and Umbrella Sampling + WHAM) provides complementary information: SMD is computationally efficient and reveals unbinding pathways, while WHAM yields converged, rigorous free energy profiles with quantified statistical uncertainties. The availability of both methods in a single pipeline enables cross-validation of ΔG_{bind} estimates, a critical practice given the known systematic biases inherent in each approach.

B. Limitations and Critical Assessment

Despite its architectural rigor, the pipeline has several important limitations that must be honestly assessed:

1) Force Field Accuracy: The AMBER ff14SB force field, while well-benchmarked for protein secondary structure and dynamics, carries systematic errors in the description of nonbonded interactions at protein-protein interfaces. Lennard-Jones parameters and partial charges are derived from gas-phase quantum calculations and empirical fitting to condensed-phase properties; their transferability to the highly heterogeneous electrostatic environment of a protein binding interface is not guaranteed. The TIP3P water model is known to overestimate water self-diffusion and underestimate viscosity, which can affect the kinetics (though typically not the thermodynamics) of binding. For high-accuracy binding free energy calculations, polarizable force fields (e.g., AMOEBA) or machine-learned potentials (e.g., ANI-2x, MACE) may be required, but at substantially higher computational cost.

2) Sampling Limitations: The non-equilibrium Jarzynski estimator is exact in the limit of infinite sampling (i.e., infinite number of pulling trajectories $N_{\mathrm{traj}} \to \infty$), but in practice suffers from severe convergence issues when the work distribution has heavy tails. The exponential average in Eq. (9) is dominated by rare, low-work trajectories that probe near-equilibrium unbinding pathways; if these trajectories are not sampled, the estimator is biased upward. This "exponential averaging problem" is particularly acute for strong binding interactions where the pulling work significantly exceeds $k_B T$. The second-order cumulant expansion (Eq. (10)) mitigates this issue but is only valid when the work distribution is approximately Gaussian—an assumption that may not hold for complex, multi-barrier unbinding landscapes.

Similarly, the WHAM reconstruction requires adequate histogram overlap between adjacent umbrella windows (enforced as IV-8 $\geq 10\%$). For systems with steep free energy barriers or complex multi-dimensional free energy surfaces projected onto a single reaction coordinate, achieving adequate overlap may require impractically narrow window spacing, greatly increasing the number of independent simulations.

3) Reaction Coordinate Choice: The use of center-of-mass distance as the sole reaction coordinate is a simplification. Protein-protein unbinding is inherently a multi-dimensional process involving translational separation, rotational reorientation, and conformational rearrangement of interface residues. Projecting this high-dimensional process onto a single scalar coordinate can introduce systematic errors in the PMF if orthogonal degrees of freedom are not equilibrated at each window [16]. More sophisticated collective variable choices—such as path-based collective variables or multi-dimensional order parameters—could improve accuracy but at the cost of increased complexity and computational expense.

4) Finite-Size Effects: Simulations of protein-protein complexes in explicit solvent with periodic boundary conditions are subject to finite-size artifacts, particularly for the long-range electrostatic interactions that dominate many binding interfaces. The minimum image convention requires that the box be large enough to prevent the protein from interacting with its periodic images (enforced as IV-7: minimum image distance $> 2r_c$). However, even when this condition is satisfied, residual finite-size corrections to the free energy can be on the order of 1–2 kcal/mol for charged protein-protein systems [13], which is comparable to the accuracy typically sought in binding free energy calculations.

5) Absence of Production-Scale Results: It must be candidly stated that the results presented in this work demonstrate pipeline *functionality and correctness* rather than production-scale scientific conclusions about the SPINK7-KLK5 binding free energy. The analytical validation (Jarzynski on harmonic potentials, WHAM on synthetic data) and the alanine dipeptide integration test confirm that the mathematical implementations are correct and that the pipeline modules interoperate correctly.

However, the production-scale SMD and Umbrella Sampling campaigns on the full SPINK7-KLK5 system—requiring hundreds of nanoseconds of aggregate simulation time on GPU-accelerated hardware—have not yet been executed. The figures shown (Figs. 1–3) are representative outputs from the visualization modules, demonstrating their capabilities with test-system data, and should not be interpreted as quantitative results for the SPINK7-KLK5 system specifically. A complete production study, including convergence analysis and comparison with experimental binding data, remains future work.

C. Comparison with Existing Tools

Several existing MD workflow tools address overlapping use cases. HTMD [18] and BioSimSpace provide high-level abstractions for MD pipeline construction, while `pmx` [19] and FESetup specialize in alchemical free energy calculations. This pipeline distinguishes itself through: (i) the enforcement of physical validity invariants as runtime assertions rather than optional post-hoc checks; (ii) the integration of both non-equilibrium (SMD/Jarzynski) and equilibrium (Umbrella/WHAM) free energy methods in a single, self-consistent framework; and (iii) the comprehensive test suite that validates not only individual modules but end-to-end pipeline integrity on a fast-executing reference system.

D. Reproducibility and Determinism

All simulation functions accept explicit random seeds to ensure exact reproducibility: NVT equilibration uses seed 42, NPT uses seed 43, production uses seed 44, and SMD replicates use base seed + replicate ID. This enables bit-for-bit reproduction of any simulation result, a critical requirement for debugging and scientific verification. The project further enforces a zero-hallucination protocol: no force field parameters are invented (all originate from published AMBER ff14SB files); no PDB identifiers are fabricated; no ΔG values are reported without corresponding simulation data; and every simulation parameter is justified by citation or established best practice.

VII. Conclusion

We have presented the design, implementation, and validation of a complete molecular dynamics simulation pipeline for computing the binding free energy of the SPINK7-KLK5 protease-antiprotease complex. The pipeline automates the full workflow from PDB retrieval and structure preparation through enhanced sampling (SMD with Jarzynski equality and Umbrella Sampling with WHAM) to thermodynamic analysis and publication-quality visualization, all built on the OpenMM engine with the AMBER ff14SB force field.

Ten physical validity invariants—enforced as runtime assertions rather than optional checks—ensure that physically meaningless results cannot propagate through the pipeline. A comprehensive test suite of 259 tests validates every module, from unit conversion utilities to end-to-end integration on an alanine dipeptide system, with all tests passing on a CPU-only platform. Analytical validation against closed-form solutions confirms the mathematical correctness of the Jarzynski estimator and WHAM solver.

The modular architecture makes the pipeline broadly applicable beyond the SPINK7-KLK5 system: any protease-inhibitor pair, drug-protein binding study, or protein-protein interaction can be investigated by substituting input structures. Key limitations—including force field accuracy, the exponential averaging problem in the Jarzynski estimator, reaction coordinate dimensionality, and the absence of production-scale results for the target system—have been identified and discussed.

Future extensions include integration of alchemical free energy perturbation methods for mutational $\Delta\Delta G$ calculations, Markov State Model construction for kinetic analysis, machine-learned force fields for improved accuracy, and GPU-accelerated production campaigns on the full SPINK7-KLK5 system to obtain the quantitative binding free energy estimates that motivated this work.

References

[1] M. E. Rothenberg, "Biology and treatment of eosinophilic esophagitis," *Gastroenterology*, vol. 137, no. 4, pp. 1238–1249, 2009.

[2] N. P. Azouz *et al.*, "The antiprotease SPINK7 serves as an inhibitory checkpoint for esophageal epithelial inflammatory responses," *Sci. Transl. Med.*, vol. 10, no. 444, p. eaap9736, 2018.

[3] M. Laskowski Jr. and I. Kato, "Protein inhibitors of proteinases," *Annu. Rev. Biochem.*, vol. 49, pp. 593–626, 1980.

[4] J. A. Maier *et al.*, "ff14SB: Improving the accuracy of protein side chain and backbone parameters from ff99SB," *J. Chem. Theory Comput.*, vol. 11, no. 8, pp. 3696–3713, 2015.

[5] W. L. Jorgensen *et al.*, "Comparison of simple potential functions for simulating liquid water," *J. Chem. Phys.*, vol. 79, no. 2, pp. 926–935, 1983.

[6] I. S. Joung and T. E. Cheatham III, "Determination of alkali and halide monovalent ion parameters for use in explicitly solvated biomolecular simulations," *J. Phys. Chem. B*, vol. 112, no. 30, pp. 9020–9041, 2008.

[7] T. Darden, D. York, and L. Pedersen, "Particle mesh Ewald: An $N \cdot \log(N)$ method for Ewald sums in large systems," *J. Chem. Phys.*, vol. 98, no. 12, pp. 10089–10092, 1993.

[8] Y. Zhang *et al.*, "A more accurate and efficient integrator for Langevin dynamics," *J. Chem. Phys.*, vol. 150, no. 12, p. 124105, 2019.

[9] J.-P. Ryckaert, G. Ciccotti, and H. J. C. Berendsen, "Numerical integration of the Cartesian equations of motion of a system with constraints," *J. Comput. Phys.*, vol. 23, no. 3, pp. 327–341, 1977.

[10] S. Izrailev *et al.*, "Steered molecular dynamics," in *Computational Molecular Dynamics: Challenges, Methods, Ideas*, pp. 39–65, Springer, 1999.

[11] C. Jarzynski, "Nonequilibrium equality for free energy differences," *Phys. Rev. Lett.*, vol. 78, no. 14, p. 2690, 1997.

[12] S. Kumar *et al.*, "THE weighted histogram analysis method for free-energy calculations on biomolecules," *J. Comput. Chem.*, vol. 13, no. 8, pp. 1011–1021, 1992.

[13] J. S. Hub, B. L. de Groot, and D. van der Spoel, "g_wham—A free weighted histogram analysis implementation including robust error and autocorrelation estimates," *J. Chem. Theory Comput.*, vol. 6, no. 12, pp. 3713–3720, 2010.

[14] S. Doudou, N. A. Burton, and R. H. Henchman, "Standard free energy of binding from a one-dimensional potential of mean force," *J. Chem. Theory Comput.*, vol. 5, no. 4, pp. 909–918, 2009.

[15] G. Schreiber and A. R. Fersht, "Energetics of protein-protein interactions: Analysis of the barnase-barstar interface by single mutations and double mutant cycles," *J. Mol. Biol.*, vol. 248, no. 2, pp. 478–486, 1995.

[16] J. D. Chodera *et al.*, "Use of the weighted histogram analysis method for the analysis of simulated and parallel tempering simulations," *J. Chem. Theory Comput.*, vol. 3, no. 1, pp. 26–41, 2007.

[17] P. Eastman *et al.*, "OpenMM 8: Molecular dynamics simulation with machine learning potentials," *J. Phys. Chem. B*, vol. 128, no. 1, pp. 109–116, 2024.

[18] S. Doerr, M. J. Harvey, F. Noé, and G. De Fabritiis, "HTMD: High-throughput molecular dynamics for molecular discovery," *J. Chem. Theory Comput.*, vol. 12, no. 4, pp. 1845–1852, 2016.

[19] V. Gapsys *et al.*, "pmx: Automated protein structure and topology generation for alchemical perturbations," *J. Comput. Chem.*, vol. 36, no. 5, pp. 348–354, 2015.